Lightning Protection Optimizer

Calculation report - geometry verification, not an IEC/EN compliance certificate

Project: 30 × 40 m demonstration

Method: rolling_sphere

LPS: III

Sphere radius (m): 45

Standard edition: IEC 62305-3:2024 (normative data not loaded)

Units: m

Geometry status: protected

Continuous geometry certificate: True

IEC/EN compliance established: No

Ground reference (m): 0.0

Geometry tolerance (m): 0.001

Sample spacing (m): 5.0

Surface certificate max depth: 8

Certificate nodes: 50

Certificate unresolved patches: 0

Certificate proof: Continuous triangle-interior bounds and full normal-cone direction bounds on physical edges/vertices; static exclusion implies inaccessibility.

Rod candidate spacing (m): 5.0

Mounting surfaces: Roof

Mount slope normal z minimum: 0.5

Objective: cost

Solver status: verification_limit

MILP gap (witness model): 0

Search method: witness-cut MILP + continuous surface certificate + independent point/direction check

Total rod cost: 400

Rod count: 4

Duration (seconds): 1.0325860540001486

Rod schedule (metres)

ID	Type	x	y	Base z	Height	Price
1	A	0	0	10	2.500	100.000
2	A	10	0	10	2.500	100.000
3	A	0	10	10	2.500	100.000
4	A	10	10	10	2.500	100.000

Object protection status

Roof: protected

Detected unprotected contact locations

Outstanding engineering checks and assumptions

Protected means continuously verified modeled geometry, not edition-wide IEC/EN compliance.

Finite candidate grid and bounded search: best certified solution found; no continuous-placement global optimality.

Static sphere-center exclusion is conservative; trapped contact centers can leave a model undetermined.

Closed externally validated solid meshes in metres; zero-radius vertical rods; infinite horizontal ground.

Bonding, down conductors, separation distances, structural checks and standard-edition applicability require engineering review.